

1. Least Squares: Three Perspectives

Let $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$. The parameters $\mathbf{x} \in \mathbb{R}^n$ produce predictions $A\mathbf{x}$. Ordinary least squares minimizes the sum of squared residuals:

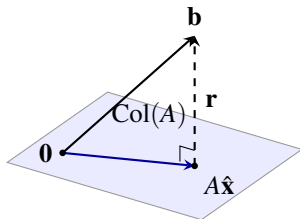
$$\hat{\mathbf{x}} \in \underset{\mathbf{x}}{\operatorname{argmin}} \|\mathbf{b} - A\mathbf{x}\|^2, \quad \hat{\mathbf{b}} = A\hat{\mathbf{x}}, \quad \mathbf{r} = \mathbf{b} - \hat{\mathbf{b}}.$$

Fitting observations. For a line $b_i \approx sa_i + c$, each row describes one observation. Matrix-vector multiplication collects the predictions:

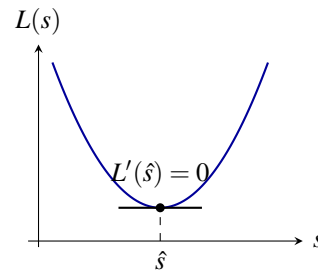
$$\underbrace{\begin{bmatrix} a_1 & 1 \\ a_2 & 1 \\ \vdots & \vdots \\ a_m & 1 \end{bmatrix}}_{\text{model matrix } A} \underbrace{\begin{bmatrix} s \\ c \end{bmatrix}}_{\text{parameters } \mathbf{x}} = \underbrace{\begin{bmatrix} sa_1 + c \\ sa_2 + c \\ \vdots \\ sa_m + c \end{bmatrix}}_{\text{predictions } A\mathbf{x}} \approx \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}}_{\text{observations } \mathbf{b}}.$$

The objective adds the squares of the vertical residuals $b_i - (sa_i + c)$ in the data plot.

Projection. The best prediction is the point in $\operatorname{Col}(A)$ closest to $\mathbf{b}$. The residual is perpendicular to this space. Here is a plane in $\mathbb{R}^3$:



Calculus. With one slope parameter, $L(s) = \|\mathbf{b} - s\mathbf{a}\|^2$ is a quadratic. For $\mathbf{a} \neq \mathbf{0}$, its minimum occurs where $L'(\hat{s}) = 0$:



In several parameters, setting each partial derivative to zero gives the same residual orthogonality condition as projection: the **normal equations**

$$A^\top (\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0} \iff A^\top A\hat{\mathbf{x}} = A^\top \mathbf{b}.$$

If the columns of A are linearly independent, the unique minimizing parameters are $\hat{\mathbf{x}} = (A^\top A)^{-1} A^\top \mathbf{b}$.

2. Building a Classifier

We would like to develop a classifier to classify points based on their distance from the origin. You are presented with the following data. Each data point $\mathbf{d}_i^T = [x_i \ y_i]^T$ has the corresponding label $l_i \in \{-1, 1\}$. For each model below, we will consider the **sign** of the output to determine if the predicted label is -1 or 1 .

x_i	y_i	l_i
-2	1	-1
-1	1	1
1	1	1
2	1	-1

Table 0: Labels for data you are classifying

- (a) You want to build a model to understand the data. You first consider a linear model, i.e. you want to find $\alpha, \beta, \gamma \in \mathbb{R}$ such that $l_i \approx \alpha x_i + \beta y_i + \gamma$. Set up a least squares problem to solve for α , β , and γ . Find all least-squares minimizers and determine whether the minimizing parameters are unique.

Answer: Rewriting the equations $\alpha x_i + \beta y_i + \gamma \approx l_i$ for $i = 1, 2, 3, 4$ in matrix form gives:

$$\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \\ x_4 & y_4 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} -2 & 1 & 1 \\ -1 & 1 & 1 \\ 1 & 1 & 1 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} \approx \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

$\mathbf{Ax} \approx \mathbf{b}$

A least-squares minimizer always exists, but the inverse formula requires linearly independent columns. Here the second and third columns are identical: since $y_i = 1$, predictions depend only on α and $\delta = \beta + \gamma$. Writing $\mathbf{u} = [-2 \ -1 \ 1 \ 2]^T$ and $\mathbf{v} = [1 \ 1 \ 1 \ 1]^T$, we have

$$\mathbf{u}^T \mathbf{v} = 0, \quad \mathbf{u}^T \mathbf{b} = \mathbf{v}^T \mathbf{b} = 0.$$

Consequently,

$$\|\mathbf{b} - \alpha \mathbf{u} - \delta \mathbf{v}\|^2 = 4 + 10\alpha^2 + 4\delta^2.$$

The minimum squared error is 4, attained exactly when $\alpha = 0$ and $\beta + \gamma = 0$. Thus all minimizing parameter vectors are

$$\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ t \\ -t \end{bmatrix}, \quad t \in \mathbb{R}.$$

The parameters are not unique.

- (b) You now consider a model with a quadratic term: $l_i \approx \alpha x_i + \beta x_i^2$ with $\alpha, \beta \in \mathbb{R}$. Set up a least squares problem to fit the model to the data. Find the least-squares minimizer and determine whether it is unique.

x_i	y_i	l_i
-2	1	-1
-1	1	1
1	1	1
2	1	-1

Table 0: Labels for data you are classifying

Answer: Rewriting the equations $\alpha x_i + \beta x_i^2 \approx l_i$ for $i = 1, 2, 3, 4$ in matrix form gives:

$$\begin{bmatrix} x_1 & x_1^2 \\ x_2 & x_2^2 \\ x_3 & x_3^2 \\ x_4 & x_4^2 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ -1 & 1 \\ 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \approx \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

$\mathbf{Ax} \approx \mathbf{b}$

The least squares solution is $\hat{\mathbf{x}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$. Here the columns of $\mathbf{A}$ are linearly independent, so $\mathbf{A}^T \mathbf{A}$ is invertible and the minimizing parameters are unique.

$$\text{We can solve for } \hat{\mathbf{x}} = \left(\begin{bmatrix} -2 & -1 & 1 & 2 \\ 4 & 1 & 1 & 4 \end{bmatrix} \begin{bmatrix} -2 & 4 \\ -1 & 1 \\ 1 & 1 \\ 2 & 4 \end{bmatrix} \right)^{-1} \begin{bmatrix} -2 & -1 & 1 & 2 \\ 4 & 1 & 1 & 4 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \end{bmatrix} = [\alpha \quad \beta]^T = \left[0 \quad \frac{-3}{17} \right]^T.$$

- (c) Finally, you consider the model: $l_i \approx \alpha x_i + \beta x_i^2 + \gamma$, where $\alpha, \beta, \gamma \in \mathbb{R}$. Independent of the work you have done so far, **would you expect this model or the model in part (b) (i.e. $l_i \approx \alpha x_i + \beta x_i^2$) to have a smaller error in fitting the data? Explain why.**

Answer: The minimum squared error cannot increase: setting $\gamma = 0$ recovers every fit available in part (b). Adding a parameter does not always strictly reduce error, but it does for these data. In fact,

$$\alpha = 0, \quad \beta = -\frac{2}{3}, \quad \gamma = \frac{5}{3}$$

produces 1 at $x_i = \pm 1$ and -1 at $x_i = \pm 2$, fitting every label exactly. The squared error is therefore zero, compared with $50/17$ in part (b). The signs also classify all four points correctly, but squared error and classification accuracy are different objectives.

3. Least Squares with Orthogonal Columns

Suppose we would like to solve the least squares problem for $\mathbf{A} \in \mathbb{R}^{3 \times 2}$ and $\mathbf{b} \in \mathbb{R}^3$; that is, find an optimal vector $\mathbf{x} \in \mathbb{R}^2$ which gets $\mathbf{A}\mathbf{x}$ closest to $\mathbf{b}$ such that the distance $\|\mathbf{e}\| = \|\mathbf{b} - \mathbf{A}\mathbf{x}\|$ is minimized. Call this optimal vector $\hat{\mathbf{x}} = \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix}$. Mathematically, we can express this as:

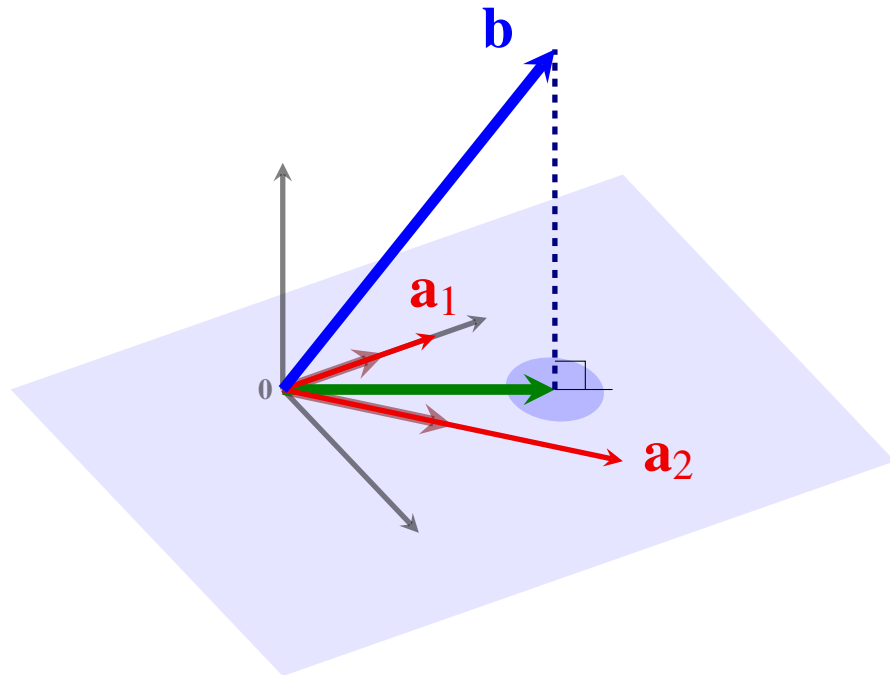
$$\|\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}\|^2 = \min_{\mathbf{x} \in \mathbb{R}^2} \|\mathbf{b} - \mathbf{A}\mathbf{x}\|^2 = \min_{\mathbf{x} \in \mathbb{R}^2} \left\| \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} - \begin{bmatrix} | & | \\ \mathbf{a}_1 & \mathbf{a}_2 \\ | & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\|^2.$$

To identify the solution $\hat{\mathbf{x}}$, we may recall the least squares formula: $\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$, which is applicable when $\mathbf{A}$ has linearly independent columns. We would now like to walk through the intuition behind this formula for the case when $\mathbf{A}$ has orthogonal columns: $\langle \mathbf{a}_1, \mathbf{a}_2 \rangle = 0$.

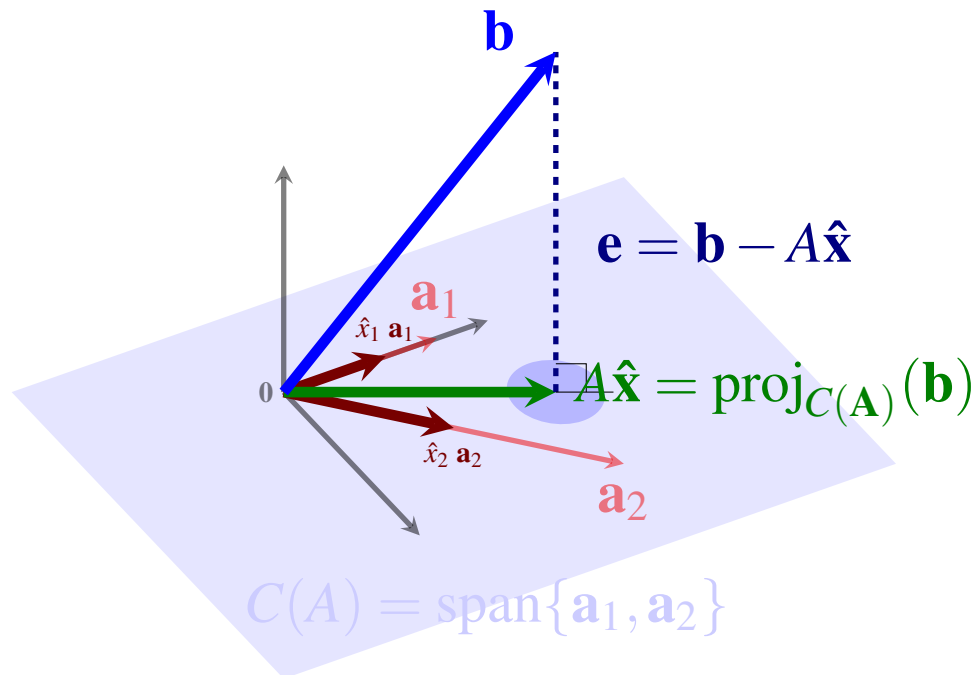
(a) On the diagram below, please label the following elements:

*NOTE: For this sub-part only, the matrix $\mathbf{A}$ does **not** have orthogonal columns.*

$\text{span}\{\mathbf{a}_1, \mathbf{a}_2\}$ $\mathbf{A}\hat{\mathbf{x}}$ $\hat{x}_1 \mathbf{a}_1$ $\hat{x}_2 \mathbf{a}_2$ $C(\mathbf{A})$ $\mathbf{e} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$ $\text{proj}_{C(\mathbf{A})}(\mathbf{b})$.



Answer:

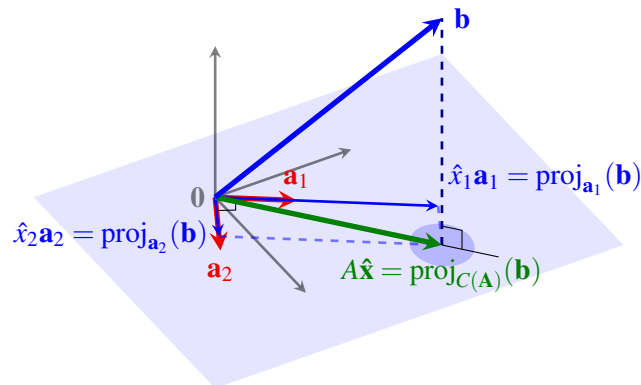


- (b) Now suppose we assume a special case of the least squares problem where the columns of $\mathbf{A}$ are orthogonal (illustrated in the figure below). Given that $\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$, and $\text{proj}_{C(\mathbf{A})}(\mathbf{b}) = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b} = \mathbf{A}\hat{\mathbf{x}}$, show the following statement holds:

$$\langle \mathbf{a}_1, \mathbf{a}_2 \rangle = 0 \implies \hat{\mathbf{x}} = \begin{bmatrix} \frac{\langle \mathbf{a}_1, \mathbf{b} \rangle}{\|\mathbf{a}_1\|^2} \\ \frac{\langle \mathbf{a}_2, \mathbf{b} \rangle}{\|\mathbf{a}_2\|^2} \end{bmatrix} \text{ and } \text{proj}_{C(\mathbf{A})}(\mathbf{b}) = \text{proj}_{\mathbf{a}_1}(\mathbf{b}) + \text{proj}_{\mathbf{a}_2}(\mathbf{b})$$

In words, the statement says that when the columns of $\mathbf{A}$ are orthogonal, the entries of the least squares solution vector $\hat{\mathbf{x}}$ can be computed by using $\mathbf{b}$ and only the single other vector $\mathbf{a}_i$, and that the projection of $\mathbf{b}$ onto $C(\mathbf{A})$ can be computed by summing the projections of $\mathbf{b}$ onto the $\mathbf{a}_i$.

RECALL... $\text{proj}_{\mathbf{a}_1}(\mathbf{b}) = \frac{\langle \mathbf{a}_1, \mathbf{b} \rangle}{\|\mathbf{a}_1\|^2} \mathbf{a}_1$, $\|\mathbf{a}\|^2 = \langle \mathbf{a}, \mathbf{a} \rangle$, $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, $\mathbf{A} = \begin{bmatrix} | & | \\ \mathbf{a}_1 & \mathbf{a}_2 \\ | & | \end{bmatrix}$



Answer: The derivation follows if we directly plug into our least-squares formula:

$$\begin{aligned}
 \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} &= \left(\begin{bmatrix} - & \mathbf{a}_1^\top & - \\ - & \mathbf{a}_2^\top & - \end{bmatrix} \begin{bmatrix} | & | \\ \mathbf{a}_1 & \mathbf{a}_2 \\ | & | \end{bmatrix} \right)^{-1} \left(\begin{bmatrix} - & \mathbf{a}_1^\top & - \\ - & \mathbf{a}_2^\top & - \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \right) \\
 &= \begin{bmatrix} \langle \mathbf{a}_1, \mathbf{a}_1 \rangle & \langle \mathbf{a}_1, \mathbf{a}_2 \rangle \\ \langle \mathbf{a}_2, \mathbf{a}_1 \rangle & \langle \mathbf{a}_2, \mathbf{a}_2 \rangle \end{bmatrix}^{-1} \begin{bmatrix} \langle \mathbf{a}_1, \mathbf{b} \rangle \\ \langle \mathbf{a}_2, \mathbf{b} \rangle \end{bmatrix} \\
 &= \begin{bmatrix} \langle \mathbf{a}_1, \mathbf{a}_1 \rangle & 0 \\ 0 & \langle \mathbf{a}_2, \mathbf{a}_2 \rangle \end{bmatrix}^{-1} \begin{bmatrix} \langle \mathbf{a}_1, \mathbf{b} \rangle \\ \langle \mathbf{a}_2, \mathbf{b} \rangle \end{bmatrix} \\
 &= \begin{bmatrix} \frac{1}{\langle \mathbf{a}_1, \mathbf{a}_1 \rangle} & 0 \\ 0 & \frac{1}{\langle \mathbf{a}_2, \mathbf{a}_2 \rangle} \end{bmatrix} \begin{bmatrix} \langle \mathbf{a}_1, \mathbf{b} \rangle \\ \langle \mathbf{a}_2, \mathbf{b} \rangle \end{bmatrix} \\
 &= \begin{bmatrix} \frac{\langle \mathbf{a}_1, \mathbf{b} \rangle}{\langle \mathbf{a}_1, \mathbf{a}_1 \rangle} \\ \frac{\langle \mathbf{a}_2, \mathbf{b} \rangle}{\langle \mathbf{a}_2, \mathbf{a}_2 \rangle} \end{bmatrix} = \begin{bmatrix} \frac{\langle \mathbf{a}_1, \mathbf{b} \rangle}{\|\mathbf{a}_1\|^2} \\ \frac{\langle \mathbf{a}_2, \mathbf{b} \rangle}{\|\mathbf{a}_2\|^2} \end{bmatrix}
 \end{aligned}$$

From this point we can quickly see that the projection onto $C(\mathbf{A})$ becomes the sum of projections onto $\mathbf{a}_1$ and $\mathbf{a}_2$ individually

$$\text{proj}_{C(\mathbf{A})}(\mathbf{b}) = \mathbf{A}\hat{\mathbf{x}} = \hat{x}_1\mathbf{a}_1 + \hat{x}_2\mathbf{a}_2 = \left(\frac{\langle \mathbf{a}_1, \mathbf{b} \rangle}{\|\mathbf{a}_1\|^2} \right) \mathbf{a}_1 + \left(\frac{\langle \mathbf{a}_2, \mathbf{b} \rangle}{\|\mathbf{a}_2\|^2} \right) \mathbf{a}_2 = \text{proj}_{\mathbf{a}_1}(\mathbf{b}) + \text{proj}_{\mathbf{a}_2}(\mathbf{b}). \quad \square$$

(c) Compute the least squares solution $\hat{\mathbf{x}} \in \mathbb{R}^2$ to the following system:

$$\min_{\mathbf{x} \in \mathbb{R}^2} \left\| \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\|^2.$$

HINT: Notice that the columns of $\mathbf{A}$ are orthogonal!

Answer: As we have shown in the previous step, there are two ways to find $\hat{\mathbf{x}}$:

- (1) Use the least-squares formula: $\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$.
- (2) Use the projection property for each term: $\hat{x}_j \mathbf{a}_j = \text{proj}_{\mathbf{a}_j}(\mathbf{b}) = \left(\frac{\langle \mathbf{a}_j, \mathbf{b} \rangle}{\langle \mathbf{a}_j, \mathbf{a}_j \rangle} \right) \mathbf{a}_j$.

IMPORTANT: This second bullet only works when the columns of $\mathbf{A}$ are orthogonal.

Since the second bullet is far easier to compute, we will employ this method here:

$$\begin{aligned}
 \hat{x}_1 &= \frac{\langle \mathbf{a}_1, \mathbf{b} \rangle}{\langle \mathbf{a}_1, \mathbf{a}_1 \rangle} = \frac{1 \cdot 1 + 0 \cdot 2 + 0 \cdot 3}{1 \cdot 1 + 0 \cdot 0 + 0 \cdot 0} = 1 \\
 \hat{x}_2 &= \frac{\langle \mathbf{a}_2, \mathbf{b} \rangle}{\langle \mathbf{a}_2, \mathbf{a}_2 \rangle} = \frac{0 \cdot 1 + 1 \cdot 2 + 1 \cdot 3}{0 \cdot 0 + 1 \cdot 1 + 1 \cdot 1} = \frac{5}{2}
 \end{aligned}$$

Thus our result is

$$\hat{\mathbf{x}} = \begin{bmatrix} 1 \\ 5/2 \end{bmatrix}. \quad \square$$